

which way from Sunday. However, most models are not so simple, and there is always the risk of model overfitting and overspecification.

By adding complexity to a model, you may make it too rigid by molding it perfectly to “predict” the past. It may then be ineffective at forecasting the future. According to López de Prado (2013), because financial data exhibits memories of various sorts, overfitting does not just lead to noise (randomness). It often leads to systematic out-of-sample losses due to mean reversion. The better the backtested results, the worse the subsequent real-time results.

There is also the problem of using the data twice, once to optimize a strategy and its parameters, and then again to judge how well the strategy predicts future returns. One needs to penalize complex or highly optimized models and to evaluate performance on fresh data not used for model development. López de Prado goes on to show that standard statistical techniques designed to prevent overfitting, such as splitting data into a development set for testing and a holdout set for cross-validation, can still be inaccurate when used for backtesting purposes. There are a number of reasons for this, the main ones being that the holdout method does not account for the number of attempted trials, and researchers may already know what happens in the second half of the data. As Nobel laureate physicist Richard Feynman pointed out, “if the process of computing the consequences is indefinite, then with a little skill any experimental result can be made to look like the expected consequences.”²

Besides multiple comparison hazards and overfitting biases, data mining can also suffer from a paucity of data. Some practitioners use only around 15 years of data to design and backtest their trading models, since that is when a number of ETFs started trading. These results are suspect, given the amount of noise there is in financial data. Return distribution parameters are time varying and regime dependent. Relatively short data samples are usually not representative of the whole, and they often do not give repeatable backtest results. The property in which every sequence or sample of data is equally representative of the whole is called ergodicity. Financial markets are most definitely nonergodic. Some stock market regimes may never repeat, while others repeat with considerably different characteristics. The following is one way to see this.

Let us define the variance ratio (VR) as the ratio of the k -period return to k times the variance of the 1-period return. When returns are uncorrelated